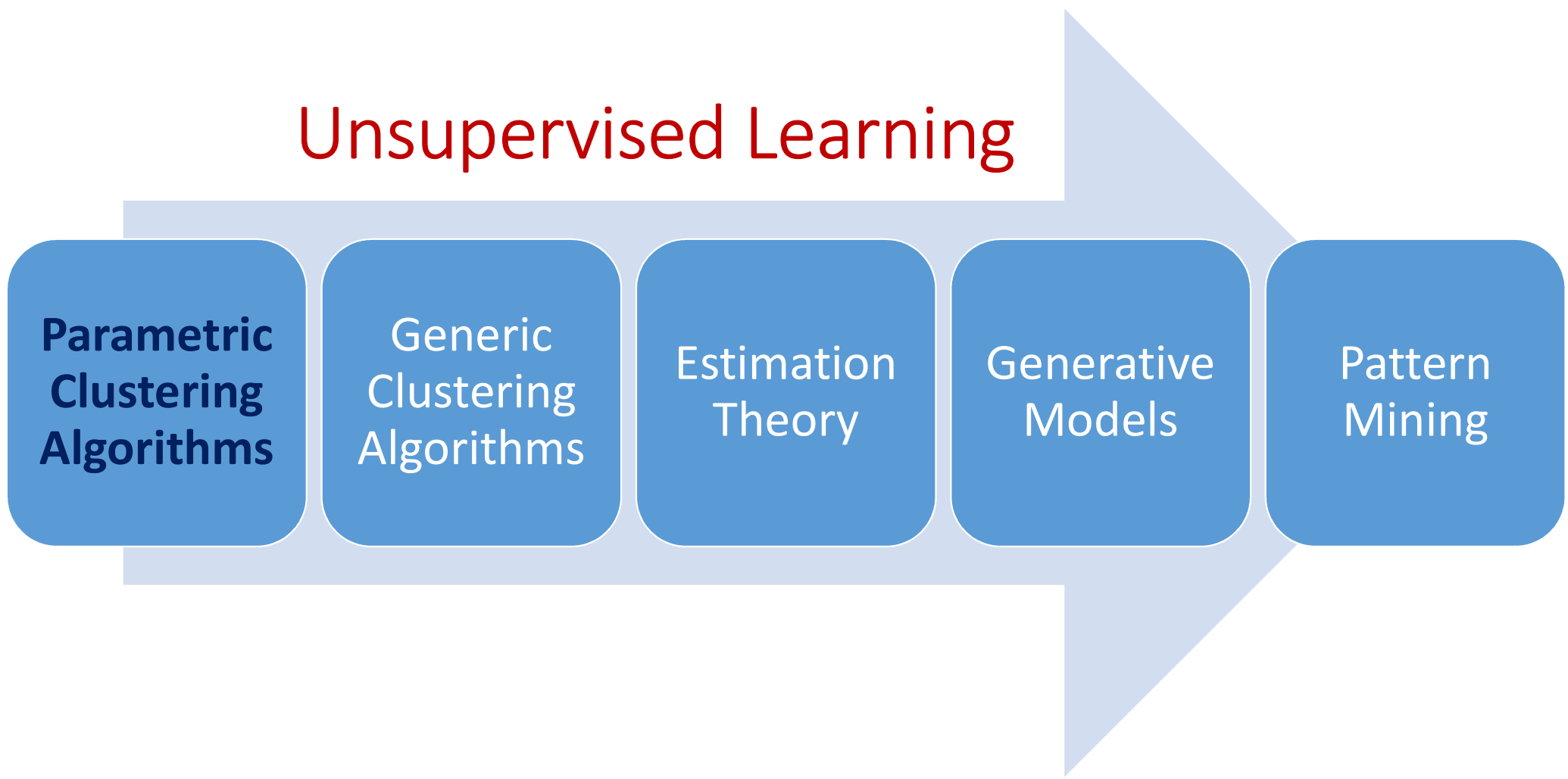


Parametric Clustering



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Unsupervised Learning



**Parametric
Clustering
Algorithms**

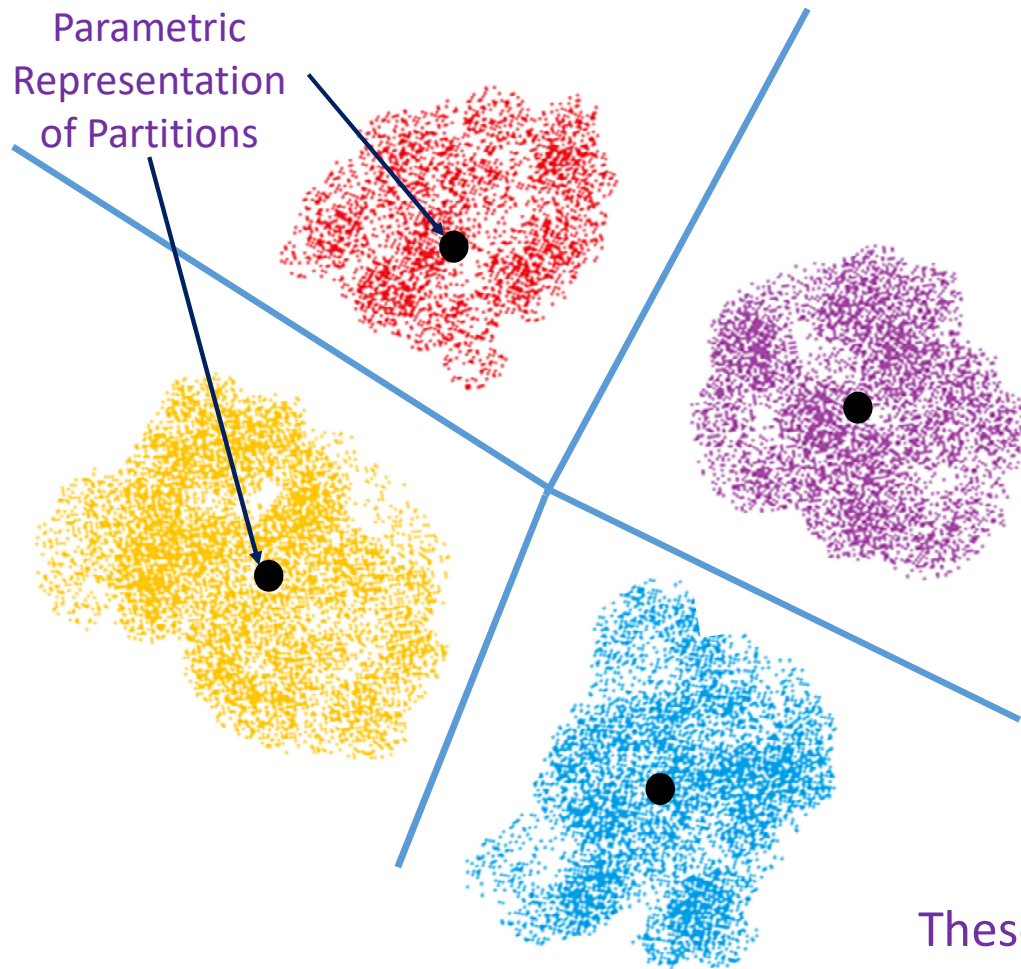
Generic
Clustering
Algorithms

Estimation
Theory

Generative
Models

Pattern
Mining

Parametric Clustering

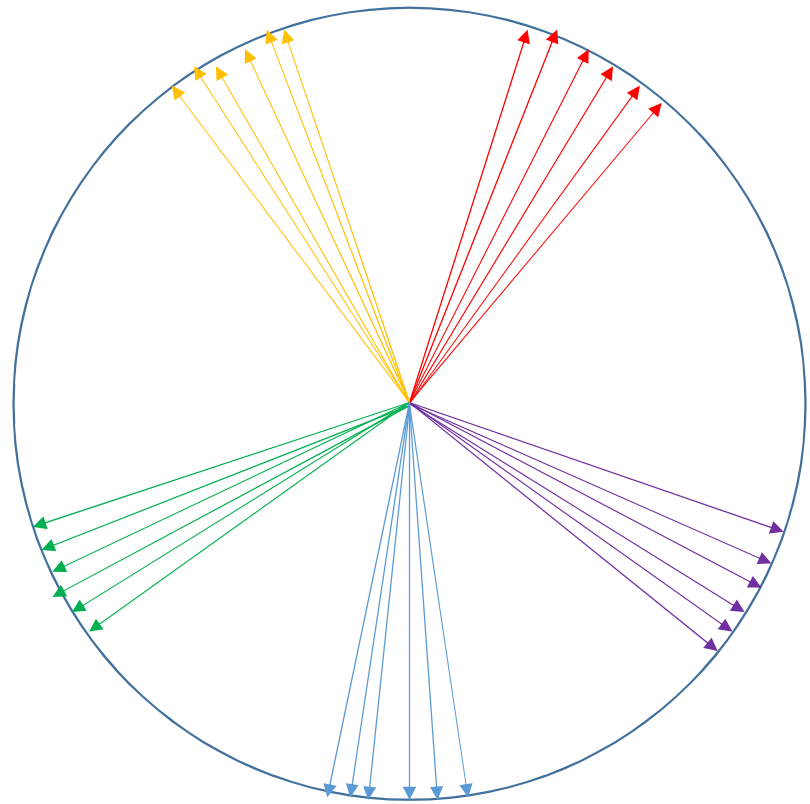


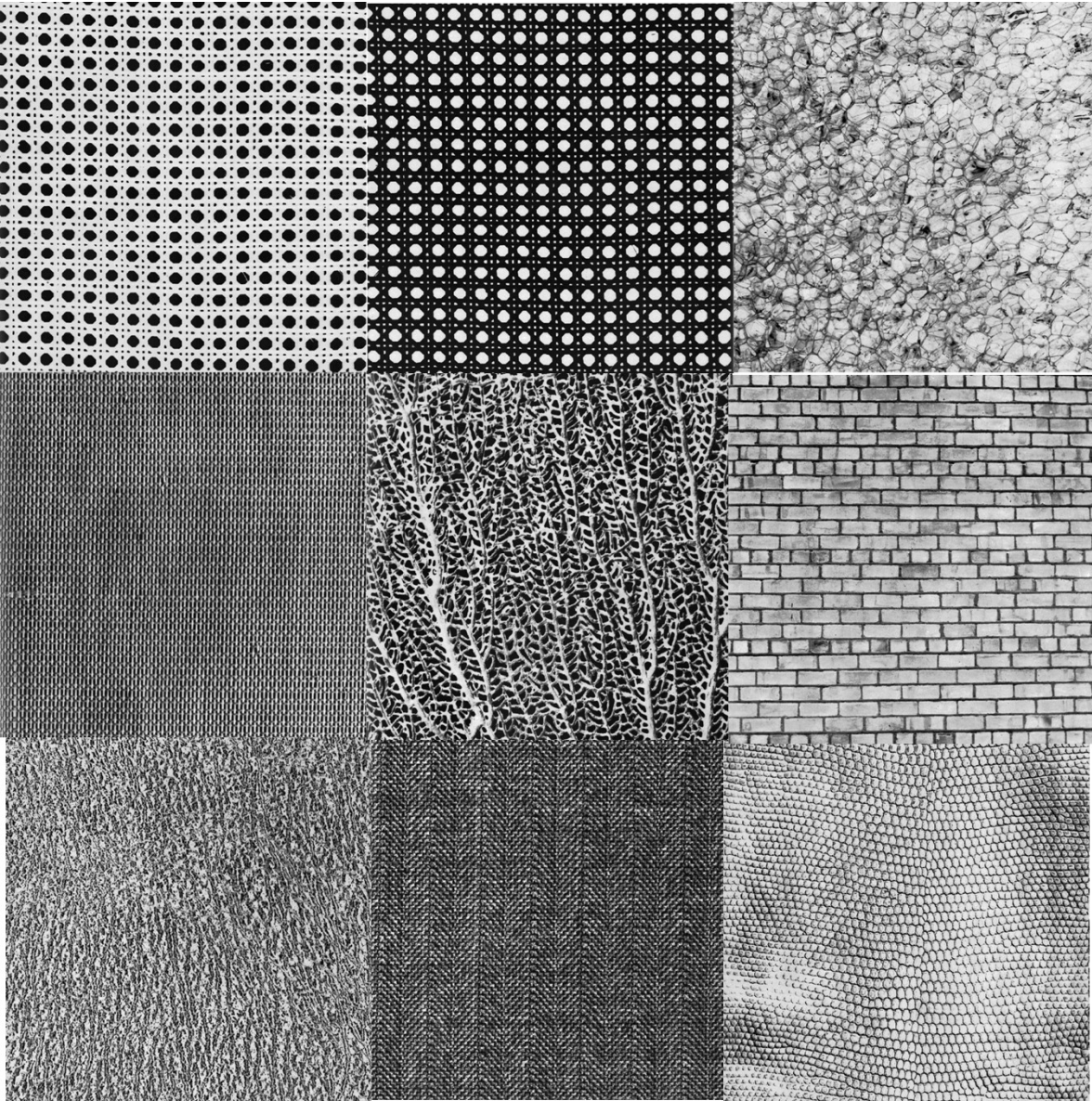
- Adaptive Thresholding
- K-Means
- Membership Weighted K-Means
- Hierarchical K-Means
- Spherical K-Means
- Incremental Clustering

These Algorithms Work on Numerical Input Data

Clustering Algorithms Partition the Input Space into (Non)Overlapping Regions

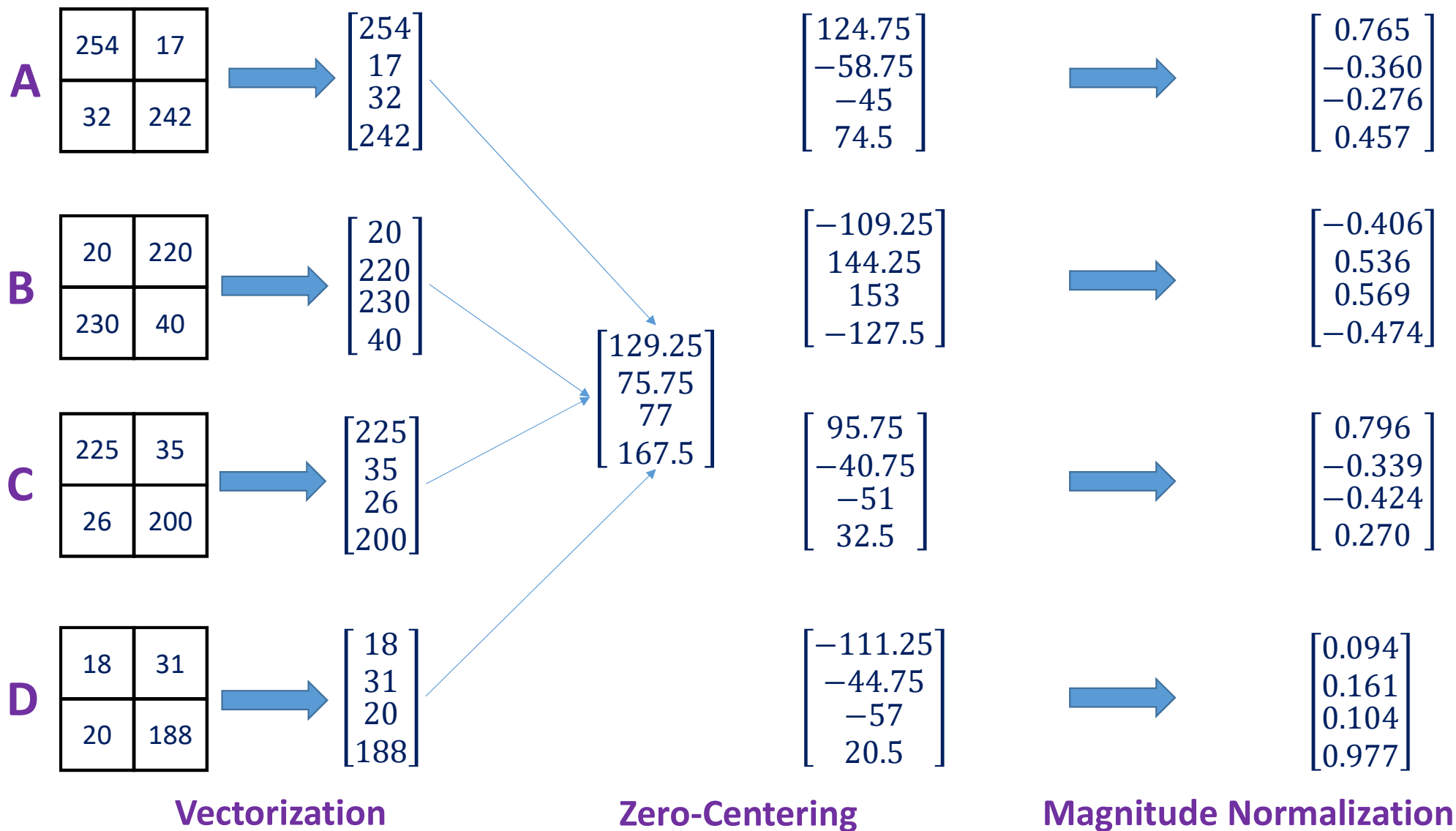
Spherical K-Means





How to Find the
Basic Patterns
whose Repetitions
Build this Images?

Can these Patterns be
Described by a Set of $m \times m$
(say 5×5) Windows?



A

254	17
32	242

 $\rightarrow$

$$\begin{bmatrix} 0.765 \\ -0.360 \\ -0.276 \\ 0.457 \end{bmatrix}$$

B

20	220
230	40

 $\rightarrow$

$$\begin{bmatrix} -0.406 \\ 0.536 \\ 0.569 \\ -0.474 \end{bmatrix}$$

C

225	35
26	200

 $\rightarrow$

$$\begin{bmatrix} 0.796 \\ -0.339 \\ -0.424 \\ 0.270 \end{bmatrix}$$

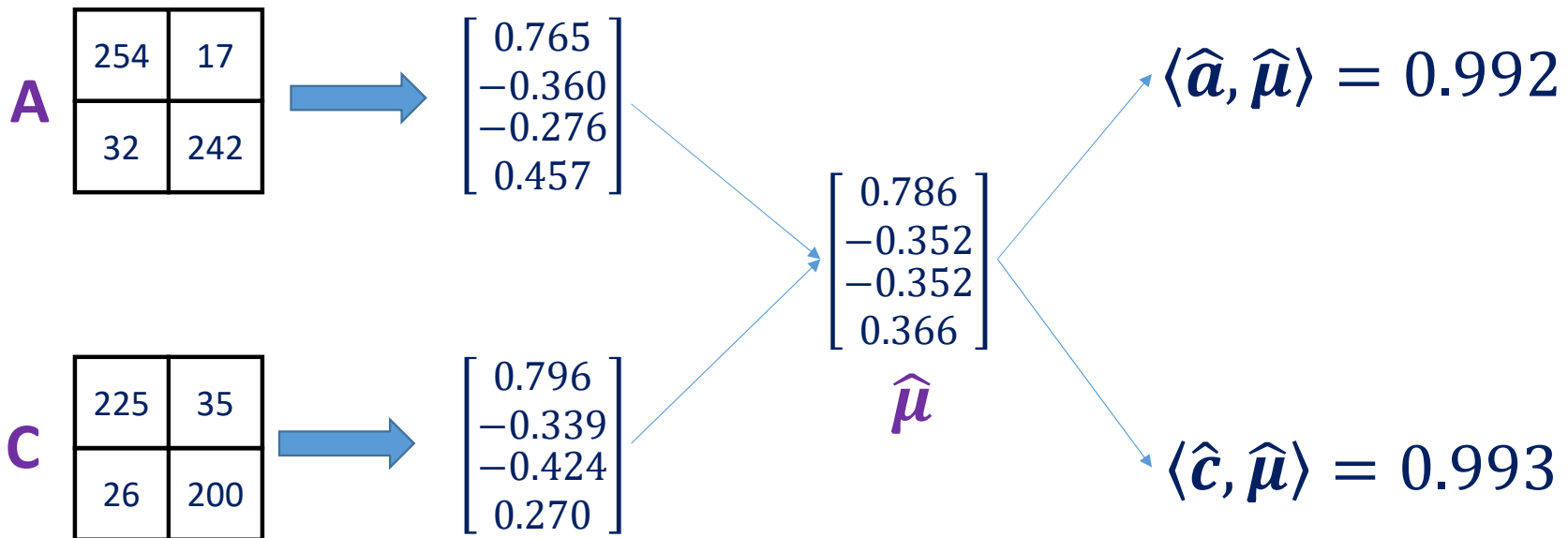
D

18	31
20	188

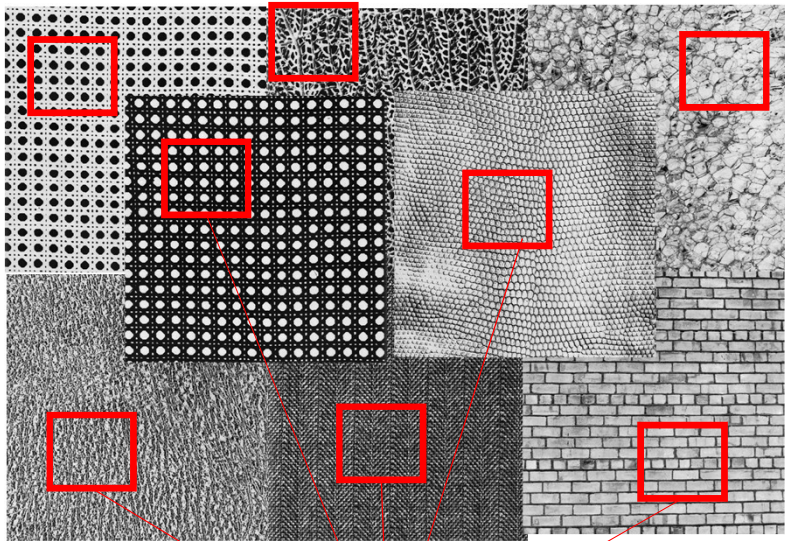
 $\rightarrow$

$$\begin{bmatrix} 0.094 \\ 0.161 \\ 0.104 \\ 0.977 \end{bmatrix}$$

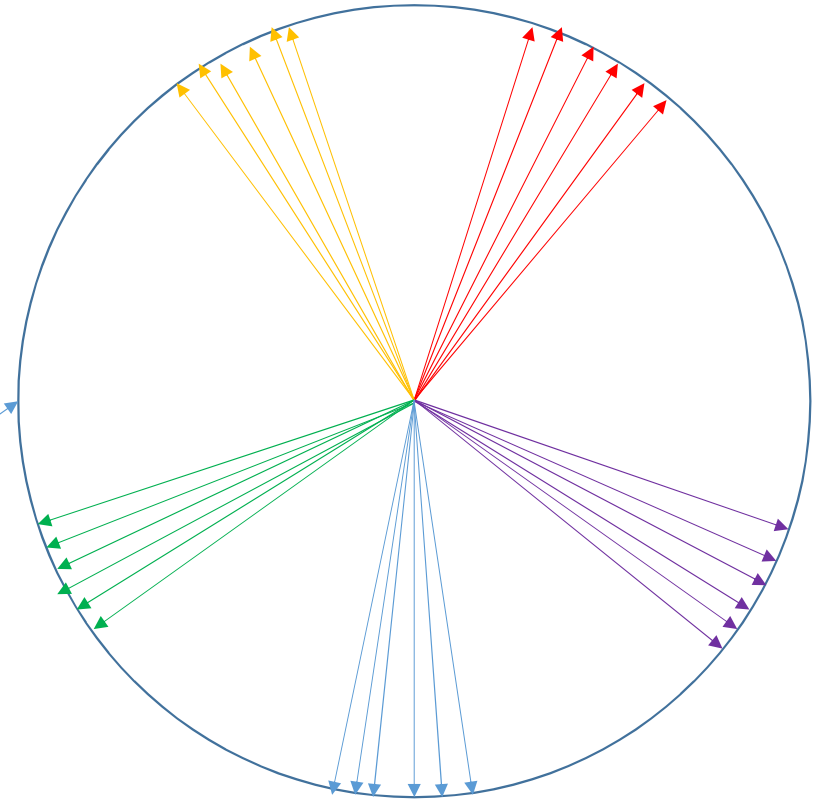
$\langle X, Y \rangle$	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	1.0	-0.877	0.971	0.431
<i>B</i>		1.0	-0.873	-0.355
<i>C</i>			1.0	0.240
<i>D</i>				1.0



Spherical K-Means



Vectorize & Magnitude
Normalize

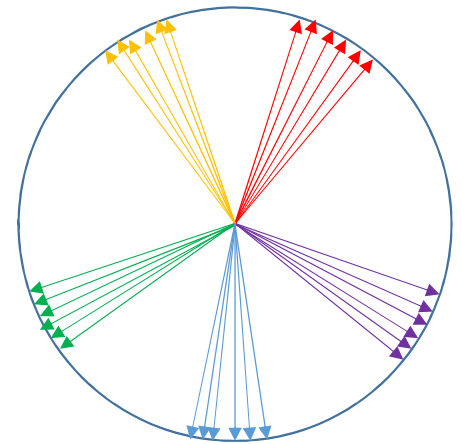


Spherical K-Means: Definitions

Input Dataset: $\tilde{\mathbf{D}} = \{\hat{\mathbf{x}}_i ; i = 1, \dots, n\}$

Spherical K-Means

$$\tilde{\mathbf{D}} = \bigcup_{j=1}^K \tilde{\mathbf{D}}_j \longrightarrow \{\hat{\boldsymbol{\mu}}_j ; j = 1, \dots, K\}$$



Spherical K-Means: Initialization

$L_i^{(\tau)}$: Label of $\hat{\mathbf{x}}_i$ at Iteration τ ($i = 1, \dots, n$)

$\hat{\boldsymbol{\mu}}_j^{(\tau)}$: Centroid of Cluster j at Iteration τ ($j = 1, \dots, K$)

Randomly Initialize K-Means using Data: $\hat{\boldsymbol{\mu}}_j^{(0)}$; $j = 1, \dots, K$

OR

Randomly Initialize N Labels to K Values: $L_i^{(0)}$; $i = 1, \dots, n$

Spherical K-Means: Algorithm

Iterations at Step τ , Labels $L_i^{(\tau-1)}$; $i = 1, \dots, n$ are available

$$\mathbf{v}_j^{(\tau)} = \frac{\sum_{i=1}^n \hat{\mathbf{x}}_i \delta \left[L_i^{(\tau-1)} - j \right]}{\sum_{i=1}^n \delta \left[L_i^{(\tau-1)} - j \right]}; \hat{\boldsymbol{\mu}}_j^{(\tau)} = \frac{\mathbf{v}_j^{(\tau)}}{\left\| \mathbf{v}_j^{(\tau)} \right\|_2}$$

Iteration: Centroid Re-estimation

Spherical K-Means: Algorithm

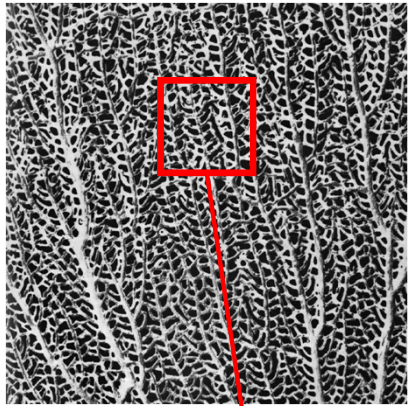
Iteration: Label Update

$$L_i^{(\tau)} = \operatorname{argmax}_{j=1,\dots,K} \left\langle \hat{\mathbf{x}}_i, \hat{\boldsymbol{\mu}}_j^{(\tau)} \right\rangle$$

Termination Condition

$$\min_{j=1,\dots,K} \left\langle \hat{\boldsymbol{\mu}}_j^{(\tau)}, \hat{\boldsymbol{\mu}}_j^{(\tau-1)} \right\rangle \geq 1 - \epsilon$$

Application: Feature Extraction



Vectorize and
Magnitude Normalize

$\hat{x}$

$$\langle \hat{x}, \hat{\mu}_1 \rangle$$

$$g_1(\cdot)$$

$$\langle \hat{x}, \hat{\mu}_j \rangle$$

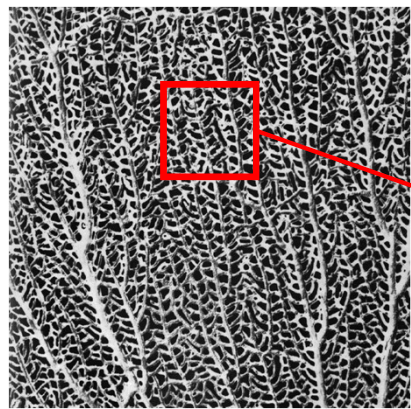
$$g_j(\cdot)$$

$$\langle \hat{x}, \hat{\mu}_K \rangle$$

$$g_K(\cdot)$$

Further
Feature
Transform

Application: Feature Extraction



$$I \rightarrow H$$



$$H[j] = \frac{1}{C_h} \sum_{\hat{x}} g_j(\langle \hat{x}, \hat{\mu}_j \rangle)$$

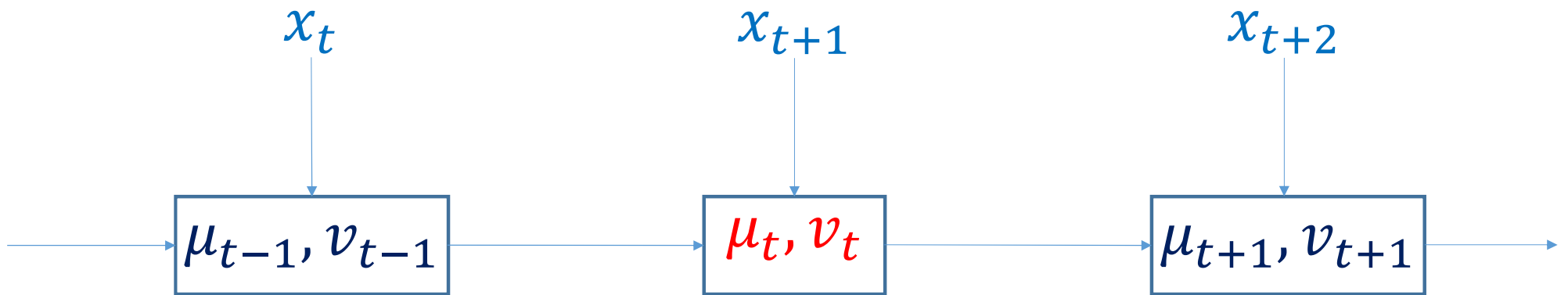
Normalization
Constant

$$C_h = \sum_{j=1}^K H[j]$$

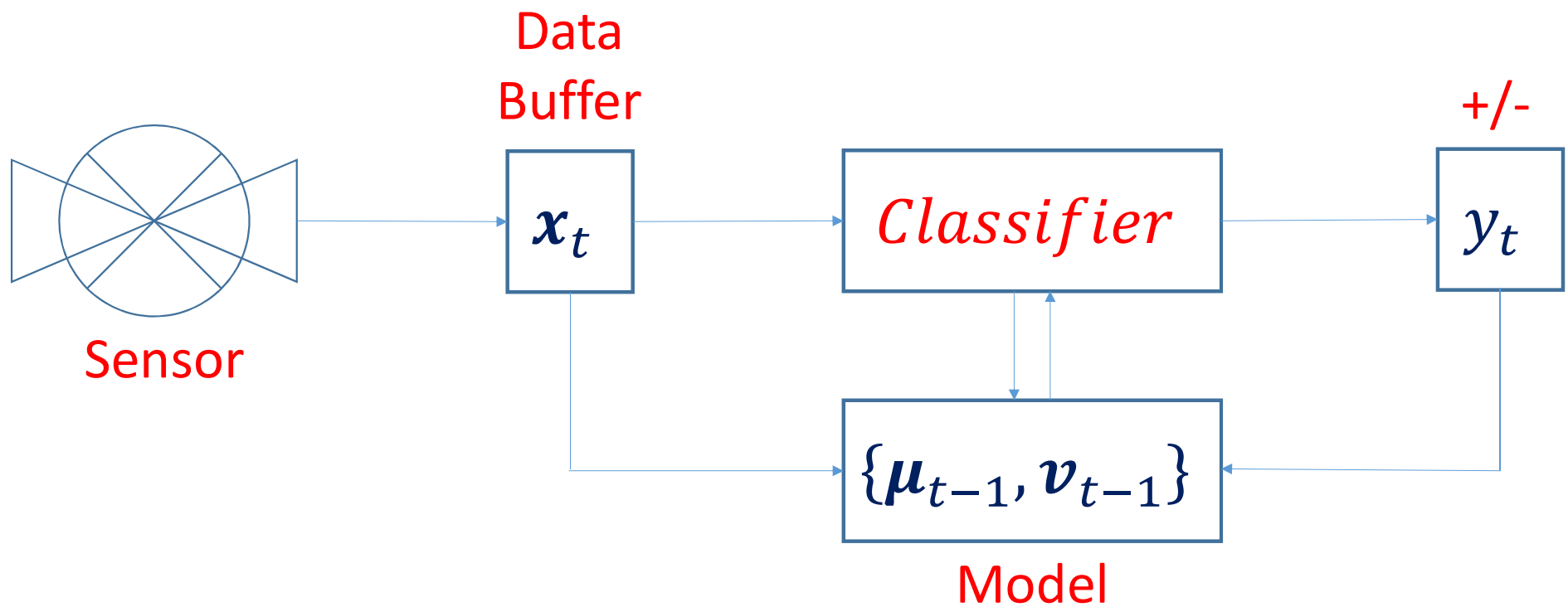
Example
Transformation

$$g_j(u) = \frac{1+u}{2}$$

Incremental Update of Mean & Variance



Application: Intrusion Detection



Definitions

x_t : Data at Instant t ($t = 1, \dots$)

μ_t : Estimated Mean of Data at Instant t ($t = 1, \dots$)

v_t : Estimated Variance of Data at Instant t ($t = 1, \dots$)

$\alpha_t = \frac{1}{t}$: Update Rate at Instant t ($t = 1, \dots$)

α : Fixed Update Rate

Incremental Mean Estimation

$$\mu_t = \frac{1}{t} \sum_{i=1}^t x_i \quad \Rightarrow \quad t\mu_t = \sum_{i=1}^{t-1} x_i + x_t = (t-1)\mu_{t-1} + x_t$$

$$t\mu_t = (t-1)\mu_{t-1} + x_t \quad \Rightarrow \quad \mu_t = \left(1 - \frac{1}{t}\right)\mu_{t-1} + \frac{1}{t}x_t$$

The Mean Update Rule

$$\mu_t = (1 - \alpha_t)\mu_{t-1} + \alpha_t x_t$$

Incremental Mean Estimation: Update Rule

μ_t is the Linear Combination of μ_{t-1} and x_t

$$\mu_t = (1 - \alpha_t)\mu_{t-1} + \alpha_t x_t$$

$\lim_{n \rightarrow \infty} \alpha_t = 0 \implies$ Mean Update Stops as t Increases

$$\alpha_t = \begin{cases} \frac{1}{t}, & t < t_c \\ \alpha, & t \geq t_c \end{cases}$$

$$\alpha = \frac{1}{t_c}$$

Incremental Mean Estimation: Exponential Forgetting

$$\mu_t = (1 - \alpha)\mu_{t-1} + \alpha x_t$$

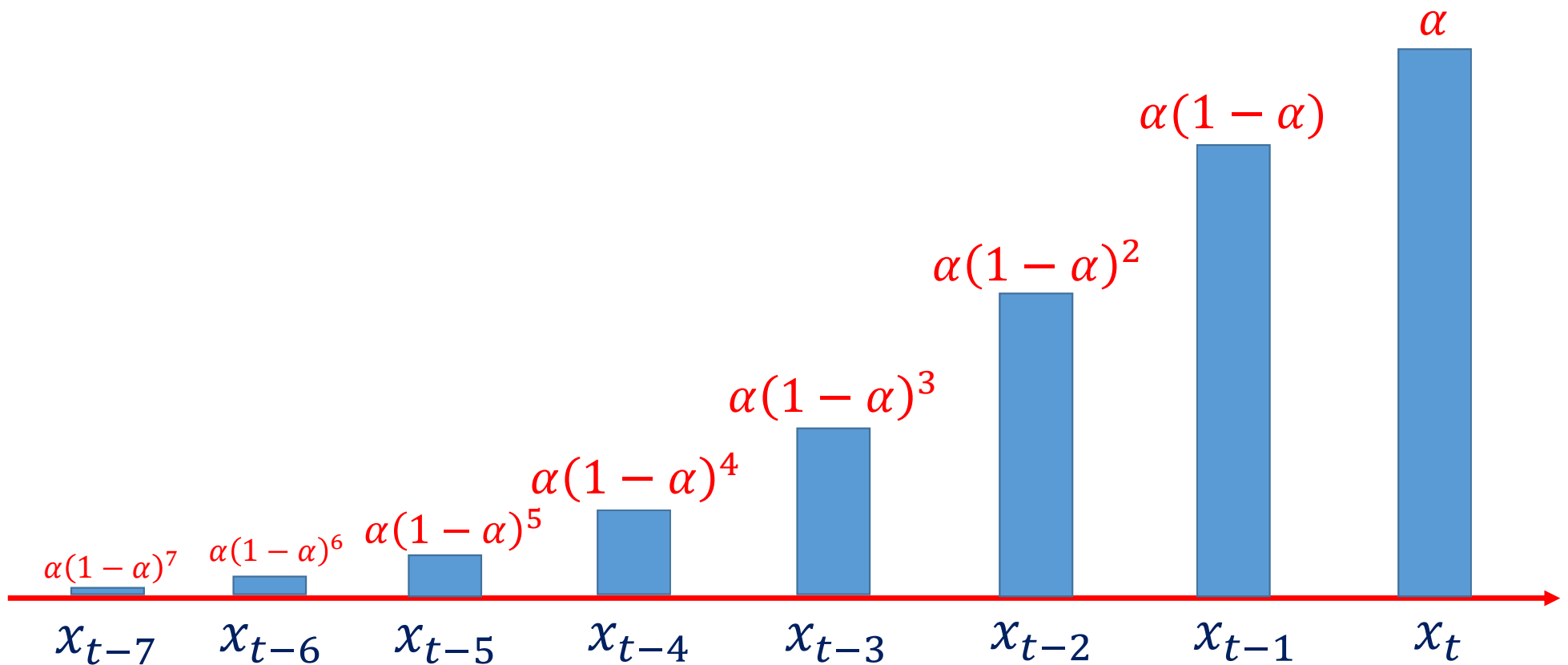
$$= (1 - \alpha)\{(1 - \alpha)\mu_{t-2} + \alpha x_{t-1}\} + \alpha x_t$$

$$= (1 - \alpha)^2\mu_{t-2} + \alpha(1 - \alpha)x_{t-1} + \alpha x_t$$

$$= (1 - \alpha)^2\{(1 - \alpha)\mu_{t-3} + \alpha x_{t-2}\} + \alpha(1 - \alpha)x_{t-1} + \alpha x_t$$

$$= (1 - \alpha)^3\mu_{t-3} + \alpha(1 - \alpha)^2x_{t-2} + \alpha(1 - \alpha)x_{t-1} + \alpha x_t$$

Incremental Mean Estimation: Exponential Forgetting



Incremental Variance Estimation

$$v_t = \frac{1}{t} \sum_{i=1}^t x_i^2 - \mu_t^2 \quad \longrightarrow \quad t(v_t + \mu_t^2) = \sum_{i=1}^{t-1} x_i^2 + x_t^2$$

$$t(v_t + \mu_t^2) = (t-1)(v_{t-1} + \mu_{t-1}^2) + x_t^2$$

$$v_t + \mu_t^2 = (1 - \alpha_t)(v_{t-1} + \mu_{t-1}^2) + \alpha_t x_t^2$$

The Variance Update Rule

$$v_t = (1 - \alpha_t)\{v_{t-1} + \alpha_t(x_t - \mu_{t-1})^2\}$$

Problem #1

Consider the problem of incremental mean and standard deviation update for streaming data problems. For a certain problem, the respective values of mean and standard deviation from the previous instant are $\mu_{t-1} = 10$ and $\sigma_{t-1} = 2$. The data at present instant is $x_t = 7$. Estimate the updated values of mean μ_t and standard deviation σ_t . The update rate is $\alpha = 0.05$.

Problem #1: Solution

$$\mu_t = (1 - \alpha)\mu_{t-1} + \alpha x_t = (1 - 0.05) * 10 + 0.05 * 7$$

$$\mu_t = 9.5 + 0.35 \longrightarrow \mu_t = 9.85$$

$$\sigma_t^2 = (1 - \alpha)\{\sigma_{t-1}^2 + \alpha(x_t - \mu_{t-1})^2\}$$

$$\sigma_t^2 = (1 - 0.05)\{2^2 + 0.05 * (7 - 10)^2\} = 0.95 * 4.45$$

$$\sigma_t^2 = 4.2275 \longrightarrow \sigma_t = 2.0561$$

Problem #2

Consider the problem of incremental mean and standard deviation update for streaming data problems. For a certain problem, the respective values of mean and standard deviation from the previous instant are μ_{t-1} and σ_{t-1} . The data at present instant is x_t . Estimate the possible values of x_t such that $\sigma_t \leq \sigma_{t-1}$. Assume a fixed update rate α .

Update Rules for Multivariate Data

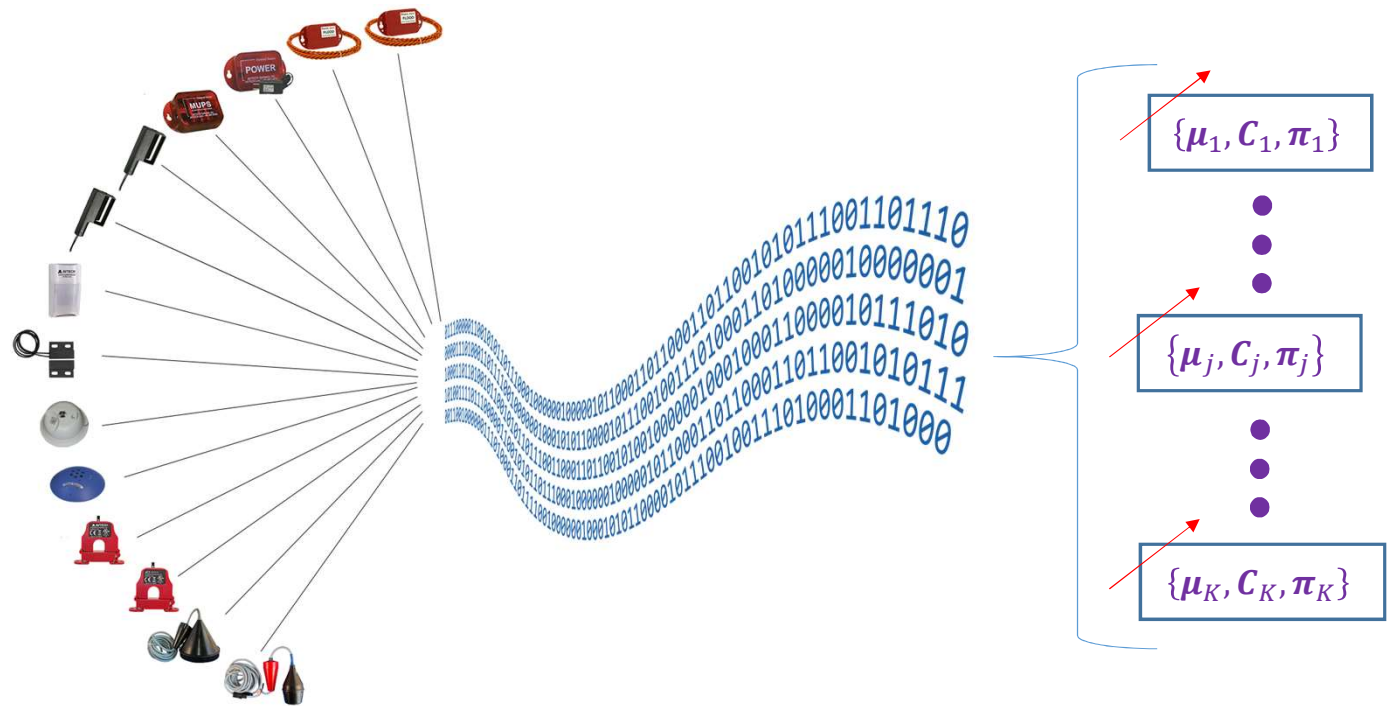
The Mean Update Rule

$$\boldsymbol{\mu}_t = (1 - \alpha_t)\boldsymbol{\mu}_{t-1} + \alpha_t\mathbf{x}_t$$

The Covariance Matrix Update Rule

$$\mathbf{C}_t = (1 - \alpha_t)\{\mathbf{C}_{t-1} + \alpha_t(\mathbf{x}_t - \boldsymbol{\mu}_{t-1})(\mathbf{x}_t - \boldsymbol{\mu}_{t-1})^T\}$$

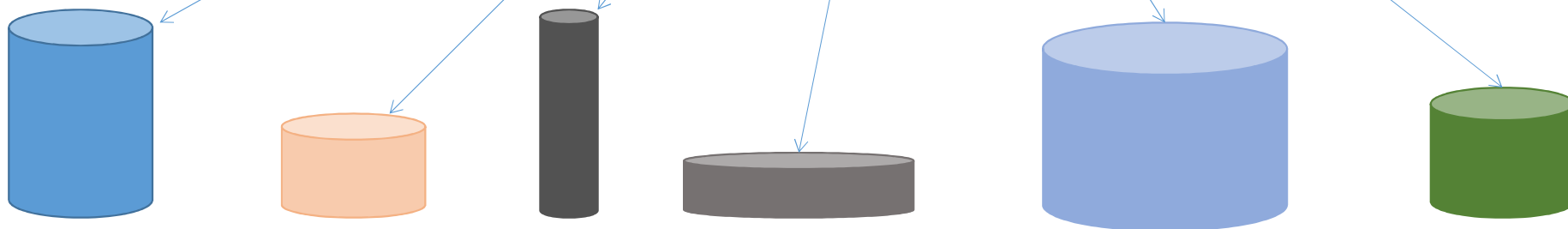
Incremental Clustering



T=1,2.....



MATCH, ASSIGN
& UPDATE



Clusters

Inc. K-Means: Definitions

x_t : Data at Instant t ($t = 1, \dots$)

$\mathbf{G}_k^{(t)}$: Cluster k at Instant t ($t = 1, \dots$)

$\mu_k^{(t)}$: Mean of $\mathbf{G}_k^{(t)}$ at Instant t ($t = 1, \dots$)

$\nu_k^{(t)}$: Variance of $\mathbf{G}_k^{(t)}$ at Instant t ($t = 1, \dots$)

$\pi_k^{(t)}$: Proportion of Data in $\mathbf{G}_k^{(t)}$ at Instant t ($t = 1, \dots$)

Inc. K-Means: Initialization

Initial
Cluster

Initial
Mean

Initial
Variance

Initial
Prior

$$\mathbf{G}_1^{(1)} = \left\{ \mu_1^{(1)} = x_1; v_1^{(1)} = v_d; \pi_1^{(1)} = 1.0 \right\}$$

Single
Cluster

Initial
Data

Default
Variance

Unity

Inc. K-Means: Cluster Set Update

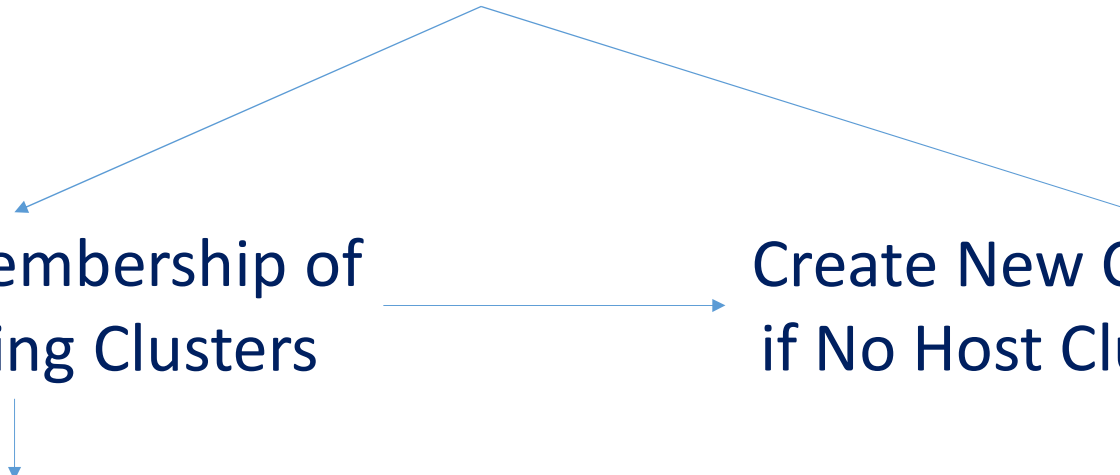
Cluster Set $\mathbf{S}_{t-1} = \{\mathbf{G}_j^{(t-1)}; j = 1, \dots, K_{t-1}\}$ available

New Data x_t arrives

Evaluate Membership of
 x_t in Existing Clusters

Create New Cluster with x_t
if No Host Cluster is Found

Update Suitable Clusters



Inc. K-Means: Hard Membership Evaluation

$$d_j^2(x_t) = \frac{\left(x_t - \mu_j^{(t-1)}\right)^2}{v_j^{(t-1)}} \quad \longrightarrow \quad J_t = \{j: d_j^2(x_t) \leq \lambda^2\}$$

IF $J_t \neq \varnothing$: Identify Host Cluster $r = \operatorname{argmin}_{j \in J} d_j^2(x_t)$

Update Host Cluster Parameters

ELSE Create New Cluster $\mathbf{G}_{K_t}^{(t)} = \{x_t; v_d; \alpha\}$

Inc. K-Means: Mean Update

IF $x_t \in \mathbf{G}_r$: Mean Update Rule

$n_r^{(t)} = t\pi_r^{(t)}$: Number of Points in $\mathbf{G}_r^{(t)}$ at Instant t

$$n_r^{(t)} \mu_r^{(t)} = (n_r^{(t)} - 1) \mu_r^{(t-1)} + x_t \Rightarrow \mu_r^{(t)} = \left(1 - \frac{1}{n_r^{(t)}}\right) \mu_r^{(t-1)} + \frac{1}{n_r^{(t)}} x_t$$

Mean Update Rule

$$\mu_r^{(t)} = \left(1 - \beta_r^{(t)}\right) \mu_r^{(t-1)} + \beta_r^{(t)} x_t$$

$$\beta_r^{(t)} = \frac{\alpha_t}{\pi_r^{(t)}}$$

Inc. K-Means: Variance Update

IF $x_t \in \mathbf{G}_r$: Variance Update Rule

$$v_r^{(t)} = \left(1 - \beta_r^{(t)}\right) \left\{ v_r^{(t-1)} + \beta_r^{(t)} \left(x_t - \mu_r^{(t-1)} \right)^2 \right\}$$

Inc. K-Means: Prior Update

IF $x_t \in \mathbf{G}_r$: Prior Update Rule

$n_r^{(t)} = t\pi_r^{(t)}$: Number of Points in $\mathbf{G}_r^{(t)}$ at Instant t

$n_r^{(t-1)} = (t-1)\pi_r^{(t-1)}$: Number of Points in $\mathbf{G}_r^{(t-1)}$ at Instant $(t-1)$

$$n_r^{(t)} = n_r^{(t-1)} + 1 \quad \longrightarrow \quad t\pi_r^{(t)} = (t-1)\pi_r^{(t-1)} + 1$$

$$\pi_k^{(t)} = (1 - \alpha_t)\pi_k^{(t-1)} + \alpha_t\delta[k - r]$$

Inc. K-Means: Soft Membership Evaluation

$$d_j^2(x_t) = \frac{\left(x_t - \mu_j^{(t-1)}\right)^2}{v_j^{(t-1)}} \quad J_t = \{j: d_j^2(x_t) \leq \lambda^2\}$$

IF $J_t \neq \varnothing$: Evaluate Membership Values $P(G_j \mid x_t)$

$$P(G_j \mid x_t) = \gamma_j^{(t)} = P(x_t \mid G_j)P(G_j) = P\left(x_t \mid \mu_j^{(t-1)}, v_j^{(t-1)}\right) \pi_j^{(t-1)}$$

Update Host Cluster Parameters

ELSE Create New Cluster $\mathbf{G}_{K_t}^{(t)} = \{x_t; v_d; \alpha\}$

Inc. K-Means: Soft Membership Evaluation

IF $J_t \neq \varphi$: Evaluate Membership Values $P(G_j \mid x_t)$

$$P(G_j \mid x_t) = \gamma_j^{(t)} = P(x_t \mid G_j)P(G_j) = P(x_t \mid \mu_j^{(t-1)}, v_j^{(t-1)}) \pi_j^{(t-1)}$$

$$\beta_j^{(t)} = \frac{\gamma_j^{(t)}}{\sum_{r \in J} \gamma_r^{(t)}}$$

$$\alpha_j^{(t)} = \frac{\alpha \gamma_j^{(t)}}{\sum_{r \in J} \gamma_r^{(t)}}$$

$$\sum_{j \in J} \beta_j^{(t)} = 1$$

$$\sum_{j \in J} \alpha_j^{(t)} = \alpha$$

Inc. K-Means: Update Rules

IF $J_t \neq \varphi$: Evaluate Soft Membership Values $P(G_j \mid x_t)$

$$\mu_j^{(t)} = \left(1 - \beta_j^{(t)}\right) \mu_j^{(t-1)} + \beta_j^{(t)} x_t$$

$$v_j^{(t)} = \left(1 - \beta_j^{(t)}\right) \left\{ v_j^{(t-1)} + \beta_j^{(t)} \left(x_t - \mu_j^{(t-1)} \right)^2 \right\}$$

$$\pi_j^{(t)} = \left(1 - \alpha_j^{(t)}\right) \pi_j^{(t-1)} + \alpha_j^{(t)}$$

ELSE Create New Cluster $\mathbf{G}_{K_t}^{(t)} = \{x_t; v_d; \alpha\}$

Inc. K-Means: Why?

- Efficient Algorithm for Streaming Data
- Provides Approximate Initial Estimation of K
- Improves Efficiency of K-Means for Big Data
- Provides Multimodal Estimation
- Good Approximation for Practical Applications

Summary

- Parametric Clustering for Numerical Data
- Spherical Clustering for Unit Vectors
- Incremental Mean & Variance Estimation
- Incremental K-Means



Thank You